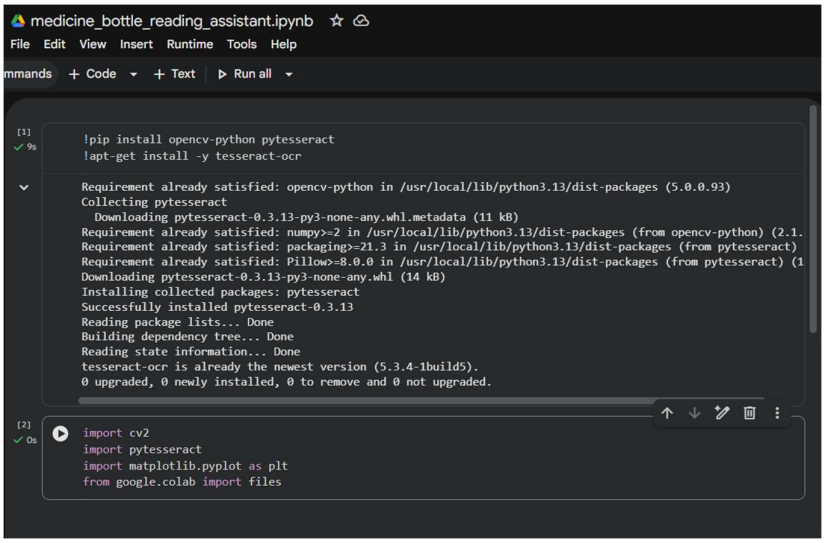
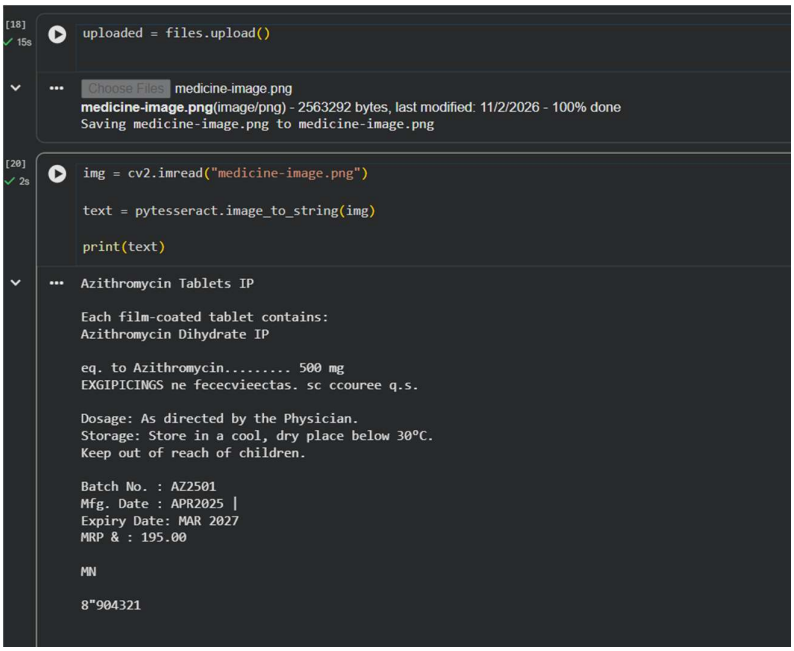
ASSIGNMENT-9 (COGNITIVE COMPUTING)

(Rahul Arora, 1024170368, 3Q33)

Q1)Medicine bottle reading assistant.(screenshots at all important steps are attached below)

Ans)








```
import re

medicine_info = {
    "Medicine Name": "Azithromycin Tablets IP",
    "Batch Number": re.search(r"Batch No\.\s*:\s*(\w+)", text).group(1),
    "Manufacturing Date": re.search(r"Mfg\.\s*Date\s*:\s*(.+)", text).group(1).strip(),
    "Expiry Date": re.search(r"Expiry Date\s*:\s*(.+)", text).group(1).strip(),
    "MRP": re.search(r"MRP\s*&?\s*:\s*([\d.]+)", text).group(1)
}

for key, value in medicine_info.items():
    print(key + ":", value)
```

```
Medicine Name: Azithromycin Tablets IP
Batch Number: AZ2501
Manufacturing Date: APR2025 |
Expiry Date: MAR 2027
MRP: 195.00
```

```
print("MEDICINE INFORMATION")
print("=" * 35)

for key, value in medicine_info.items():
    print(key + ":", value)

print("=" * 35)
print("Dosage: As directed by the physician.")
print("Storage: Store in a cool, dry place below 30°C.")
print("OCR processing completed.")
```

```
MEDICINE INFORMATION
===================================
Medicine Name: Azithromycin Tablets IP
Batch Number: AZ2501
Manufacturing Date: APR2025 |
Expiry Date: MAR 2027
MRP: 195.00
===================================
Dosage: As directed by the physician.
Storage: Store in a cool, dry place below 30°C.
OCR processing completed.
```

```
from datetime import datetime

expiry = medicine_info["Expiry Date"]

expiry_date = datetime.strptime(expiry, "%b %Y")

today = datetime.now()

if expiry_date < today:
    print("Warning: This medicine has expired!")
else:
    print("This medicine is not expired.")
```

```
This medicine is not expired.
```

```
print("MEDICINE READING ASSISTANT")
print("=" * 40)

for key, value in medicine_info.items():
    print(key + ":", value)

print("-" * 40)

if expiry_date < today:
    print("STATUS: EXPIRED")
else:
    print("STATUS: NOT EXPIRED")

print("=" * 40)
print("Please verify the information on the original label.")
```

```
MEDICINE READING ASSISTANT
========================================
Medicine Name: Azithromycin Tablets IP
Batch Number: AZ2501
Manufacturing Date: APR2025 |
Expiry Date: MAR 2027
MRP: 195.00
----------------------------------------
STATUS: NOT EXPIRED
========================================
Please verify the information on the original label.
```

Test image -



Prediction –

```
medicine_assistant()

Choose Files medicine-image2.png
medicine-image2.png(image/png) - 2628981 bytes, last modified: 11/2/2026 - 100% done
Saving medicine-image2.png to medicine-image2.png

MEDICINE READING ASSISTANT
==================================
Extracted Medicine Information:
Rx
Amoxicillin and
Potassium Clavulanate

Batch No. ; AC2487
Tablets IP Mfg. Date : JUN 2025

Augmentin® 625 Expiry Date : MAY 2027

Each film-coated tablet contains : MRP & : 248.00 |
Amoxicillin Trinydrate IP (Incl. of all taxes)
eq. to Amoxicillin ..... 500 mg

Potassium Clavulanate Diluted IP
eq. to Clavulanic Acid .... 125 mg
Dosage : As directed by the Physician.

Storage : Store below 25°C in a dry place.
Protect from moisture.

Keep out of reach of children.

901296 045678


==================================
Expiry date information detected.
Please verify the information on the original label.
```

Q1 complete

Q2) Emotion-aware Cognitive Assistant using Facial Expression Detection.

Ans)

```
!pip install transformers torch pillow
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.13/dist-packages (5.21.0)
Requirement already satisfied: torch in /usr/local/lib/python3.13/dist-packages (2.11.0+cpu)
Requirement already satisfied: pillow in /usr/local/lib/python3.13/dist-packages (11.3.0)
Requirement already satisfied: huggingface-hub<3.0,>=1.31.0 in /usr/local/lib/python3.13/dist-packages (from transformers) (1.33.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.13/dist-packages (from transformers) (2.1.3)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from transformers) (26.3)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.13/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.13/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<0.24.0,>=0.23.1 in /usr/local/lib/python3.13/dist-packages (from transformers) (0.23.3)
Requirement already satisfied: typer in /usr/local/lib/python3.13/dist-packages (from transformers) (0.27.3)
Requirement already satisfied: safetensors>=0.8.0 in /usr/local/lib/python3.13/dist-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.60 in /usr/local/lib/python3.13/dist-packages (from transformers) (4.67.2)
Requirement already satisfied: click<9.0.0,>=8.4.2 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (8.5.0)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (4.2.4)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (2025.12.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (1.7.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (0.28.1)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (4.16.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0->transformers) (4.15.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0->transformers) (2026.10.23)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0->transformers) (3.20)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0->transformers) (0.16.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.13/dist-packages (from torch) (80.10.2)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.13/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.13/dist-packages (from torch) (3.7)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from torch) (3.1.6)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->torch) (3.0.4)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.13/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.13/dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.13/dist-packages (from typer->transformers) (0.0.5)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer->transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer->transformers) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)
```

```
import cv2
import torch
from PIL import Image
from transformers import pipeline
from google.colab import files
import matplotlib.pyplot as plt
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config.json: 100% 915/915 [00:00<00:00, 85.7kB/s]
model.safetensors: reconstructing file: 100% 343MB / 343MB, 26.9MB/s
model.safetensors: downloading bytes: 325MB, 29.3MB/s
Loading weights: 100% 200/200 [00:00<00:00, 3283.48it/s]
preprocessor_config.json: 100% 228/228 [00:00<00:00, 21.0kB/s]
Emotion detection model loaded successfully!
```

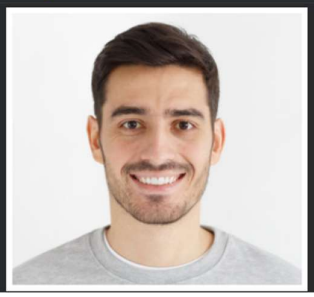
```
uploaded = files.upload()
```

```
Choose Files face.png
face.png(image/png) - 111936 bytes, last modified: 11/2/2026 - 100% done
Saving face.png to face.png
```

```
filename = list(uploaded.keys())[0]

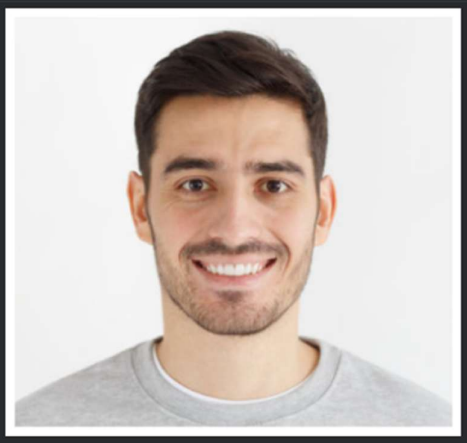
img = Image.open(filename).convert("RGB")

plt.figure(figsize=(6, 6))
plt.imshow(img)
plt.axis("off")
plt.show()
```



```
img = Image.open(filename).convert("RGB")

plt.figure(figsize=(6, 6))
plt.imshow(img)
plt.axis("off")
plt.show()
```



```
results = emotion_model(img)

print("Detected Emotion:", results[0]["label"])
print("Confidence:", round(results[0]["score"] * 100, 2), "%")
```

```
Detected Emotion: happy
Confidence: 98.32 %
```

```
emotion = results[0]["label"].lower()

responses = {
    "happy": "You seem cheerful today! Keep smiling.",
    "sad": "You seem a little down. Take some time for yourself.",
    "angry": "You seem frustrated. Take a moment to relax.",
    "fear": "You seem worried. Try to stay calm.",
    "surprise": "Something seems to have caught your attention!",
    "disgust": "You seem displeased. Take a moment to reset.",
    "neutral": "You seem calm and composed."
}

print("Detected Emotion:", emotion)
print("Assistant Response:", responses.get(emotion, "Thank you for sharing your expression!"))
```

```
Detected Emotion: happy
Assistant Response: You seem cheerful today! Keep smiling.
```

```
def emotion_assistant():
    uploaded = files.upload()

    for filename in uploaded.keys():
        img = Image.open(filename).convert("RGB")

        results = emotion_model(img)

        emotion = results[0]["label"].lower()
        confidence = round(results[0]["score"] * 100, 2)

        responses = {
            "happy": "You seem cheerful today! Keep smiling.",
            "sad": "You seem a little down. Take some time for yourself.",
            "angry": "You seem frustrated. Take a moment to relax.",
            "fear": "You seem worried. Try to stay calm.",
            "surprise": "Something seems to have caught your attention!",
            "disgust": "You seem displeased. Take a moment to reset.",
            "neutral": "You seem calm and composed."
        }

        print("\nEMOTION-AWARE ASSISTANT")
        print("=" * 35)
        print("Detected Emotion:", emotion)
        print("Confidence:", confidence, "%")
        print("Assistant Response:", responses.get(emotion, "Thank you!"))
        print("=" * 35)
```

```
print("=" * 35)
```

```
emotion_assistant()
```

```
Choose Files  Screenshot ...223014.png
Screenshot 2026-11-02 223014.png(image/png) - 125160 bytes, last modified: 11/2/2026 - 100% done
Saving Screenshot 2026-11-02 223014.png to Screenshot 2026-11-02 223014.png

EMOTION-AWARE ASSISTANT
===================================
Detected Emotion: neutral
Confidence: 99.06 %
Assistant Response: You seem calm and composed.
===================================
```